

- * electrical impulses travel along neurones
- ↳ neurones are nerve cells

Mammalian Nervous System :-

- central nervous system (CNS) consists of the brain and the spinal cord
- peripheral nervous system (PNS) consists of all of the nerves in the body

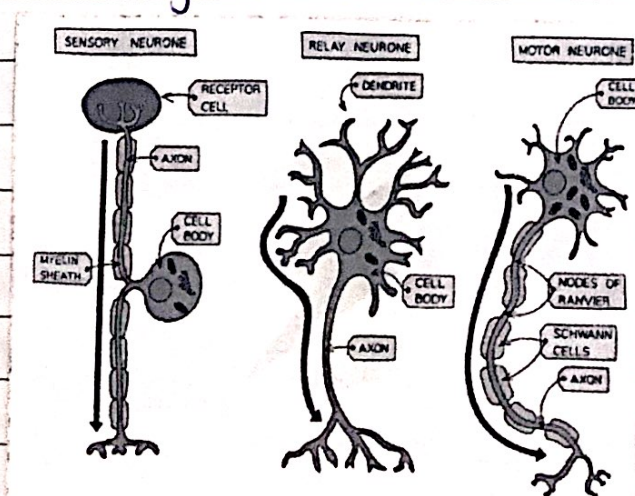
→ the nervous system coordinates and regulates body functions

3 main types of neurones :-

sensory neurones carry impulses from sense organs to the CNS (brain or spinal cord)

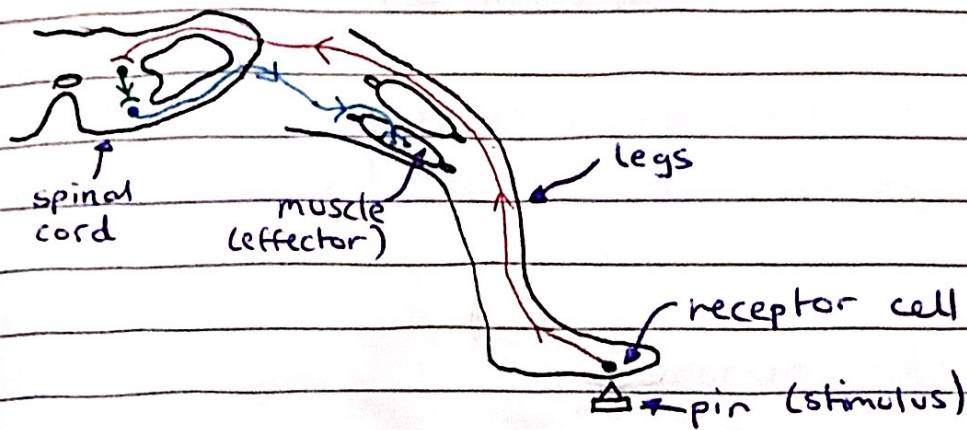
relay neurones are found inside the CNS & they connect sensory and motor neurones

motor neurones carry impulses from the CNS to effectors (an organ/cell that reacts to stimulus)



- receptors are specialised cells that detect a change in the environment and stimulate electrical impulses in response.

- relay
- motor



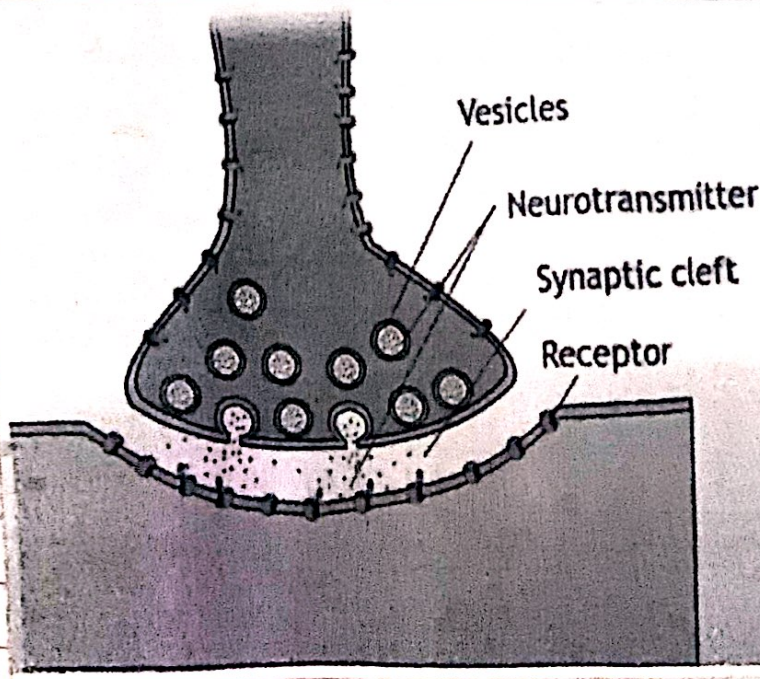
stimulus → receptor → sensory → relay → motor → effector → response
 ↳ as a result, the muscle contracts.

reflex action:

a means of automatically & rapidly integrating and coordinating stimuli with the responses of effectors.

- a synapse is a junction between two neurones

Synapse::



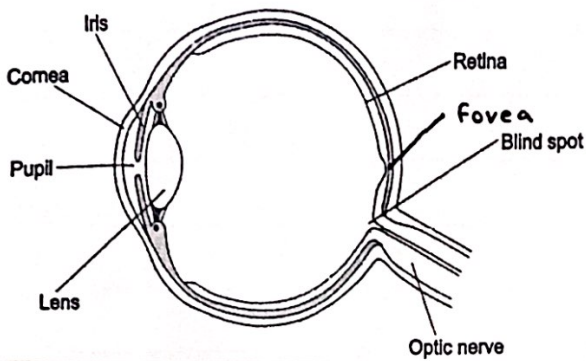
- an impulse stimulates the release of neurotransmitter into the synaptic gap
- the neurotransmitter diffuse across the gap
- the neurotransmitter bind with the receptor proteins in the next neurone
- an impulse is then stimulated in the next neurone.

• synapses ensure that impulses travel in one direction (only)

Sense organs:

groups of receptor cells responding to specific stimuli
e.g. light, sound, touch, temperature, chemicals

Eye:-



cornea: reflects light

iris: controls amount of light that enters the pupil

lens: focuses light on to the retina

retina: contains light receptors, some sensitive to light of different colors

optic nerve: carries impulses to the brain

blind spot: the point where the optic nerve joins the retina, there are no light-sensitive rods/cones, so the light falling there will not result in an image, instead the brain 'fills in' that spot. The blind spot is where we cannot detect an object in our peripheral vision even if it's there.

Pupil reflex:-

- a reflex action carried out to protect the retina
- in dim light, pupils dilate (widen) in order to allow as much light into the eye as possible
- in bright light, pupils constrict (narrow) in order to prevent too much light from entering the retina and damaging it.

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CARCD

in pupil reflex, a group of antagonistic muscle groups act together (in the iris)

the radial & circular muscles work antagonistically

↳ when one contracts, the other relaxes

dim light = radial contract & circular relax

bright light = radial relax & circular contract

Accommodation:

- the way the lens brings about fine focusing
- the lens is elastic so it changes shape depending on whether the suspensory ligaments tighten or loosen.
- this change is brought by the contraction & relaxation of the ciliary muscles.

- when an object is close up

↳ ciliary muscles contract

↳ suspensory ligaments loosen

↳ suspensory ligaments stop pulling on lens

↳ lens becomes fatter

↳ more refraction

- when an object is far away

↳ ciliary muscles relax

↳ suspensory ligaments tighten

↳ suspensory ligaments pull on the lens

↳ lens becomes thinner

↳ less refraction.

Rods & Cones ::

- rods are sensitive to dim light

cones distinguish between different colors in bright light (R, G, B)

- the fovea is an area on the retina where almost all of the cone cells are found
- rod cells are found all over the retina except the blind spot.

* rods have greater sensitivity for night vision

* there are 3 types of cones that absorb light of different colors for color vision

HORMONES

17/01/23

Hormones::

- a chemical substance produced by a gland and carried by the blood, which alters the activity of one or more specific target organs.

adrenal glands → adrenaline

(found on top of the kidneys)

pancreas → insulin & glucagon

testes → testosterone (in male body)

ovaries → oestrogen (in female body)

- secreted in 'fight or flight' situations::

- breathing rate ↑

- heart rate ↑

- pupil diameter ↑

* controls metabolic activity by increasing heart rate and the blood glucose concentration.

HOMEOSTASIS

17/01/23

Homeostasis ::

- the maintenance of a constant internal environment
- * insulin decreases blood glucose concentration

Homeostatic control :-

- if the level of something rises, control systems are turned on to reduce it
- if the level of something falls, control systems are turned on to raise it
- * Negative feedback is a mechanism that controls and responds when conditions change from the set point and returns them to the set point.

① when blood glucose rises

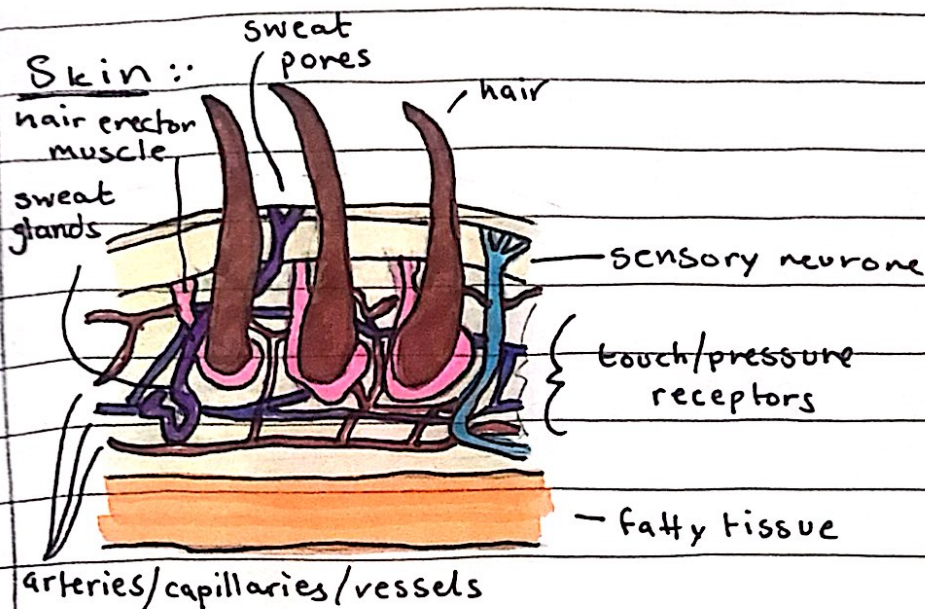
- ↳ insulin is produced
- ↳ insulin stimulates liver/muscle cells
- ↳ converts excess glucose into glycogen

② when blood glucose falls

- ↳ glucagon is produced
- ↳ glucagon stimulates liver/muscle cells
- ↳ converts stored glycogen into glucose

Type 1 diabetes ::

- condition where blood glucose levels are far too high because the cells in the pancreas are unable to produce insulin
- can be treated by injecting insulin
- symptoms = extreme thirst, weakness, tiredness or blurred vision
- control by good diet & by exercising.



- maintenance of a constant internal body temperature is done by :-

* insulation

* sweating

* shivering

↳ this is done by the temp. receptors in the skin that send nerve impulses to the brain via sensory neurones.

Vasoconstriction

- when we are cold, blood flow in capillaries slows down because the arterioles leading to the capillaries get narrower.
- This reduces the amount of heat lost from blood by radiation since there is less blood flow at the surface

Vasodilation

- when we are hot, blood flow in capillaries increases because blood vessels leading to the capillaries get wider.
- This cools the body as the blood flows at a faster rate so more heat is lost by radiation.

TROPIC RESPONSES

17/01/23

Gravitropism:

- a response in which parts of a plant grow towards or away from gravity

Phototropism:

- a response in which parts of a plant grow towards or away from the direction of a light source

Investigate

phototropism:-

- (A) in front of a unilateral light ↗ light-proof box
- (B) in front of a unilateral light + on a clinostat
- (C) in complete darkness

- ↳ (A) will grow towards the light source
- ↳ (B) will grow straight up (shoots were evenly exposed to light due to clinostat)
- ↳ (C) will grow straight up (looking for light & will be tall and thin with yellowing leaves)

gravitropism:-

- (A) facing downwards
 - (B) horizontally
 - (C) facing upwards
- (in two petri dishes, on damp cotton wool where one dish is on its side and the other one is too but on a clinostat & both are in a light-proof box so no phototropism influences the seedlings)

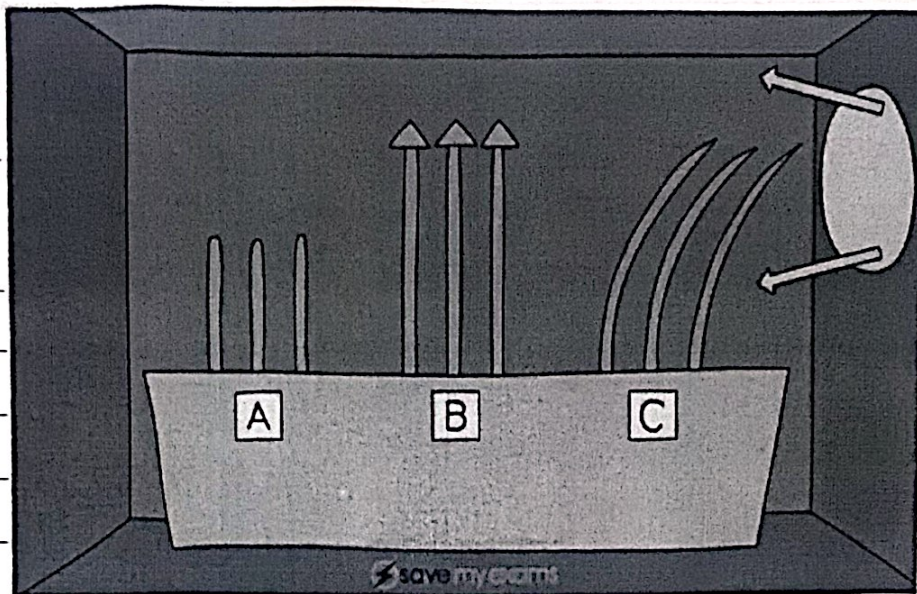
- ↳ (PD1) all roots grow downwards regardless of initial position
- ↳ (PD2) roots grow neither up nor down (cancelled out the effect of gravity via the clinostat)

* in PD1 the shoots grow upwards while in PD2 the shoots grow neither up nor down

→ phototropism & gravitropism of a shoot are examples of chemical control of a plant's growth

Auxin:.

- is a growth hormone
- made in the shoot tip
- diffuses through the plant from the shoot tip
- unequally distributed in response to light/gravity
- stimulates cell elongation



Investigating the phototropic response set-up

Investigating the phototropic response results:

	SEEDLING A	SEEDLING B	SEEDLING C
TREATMENT	THE TIPS OF THE STEMS HAVE BEEN REMOVED	NO LIGHT REACHES THE TIPS	MORE LIGHT REACHES ONE SIDE OF THE TIPS
EFFECT ON AUXIN CONCENTRATION	NO AUXIN IS PRODUCED	EQUAL CONCENTRATION OF AUXIN ON BOTH SIDES OF THE TIP	GREATER CONCENTRATION OF AUXIN ON SHADED SIDE
RESULT	THE STEMS DO NOT GROW LONGER	THE STEMS GROW EVENLY AND LONGER ON BOTH SIDES	THE CELLS ON THE DARKER SIDE OF THE STEMS GROW LONGER AND FASTER THAN CELLS IN THE LIGHT
REASON	THIS PROVES THAT AUXIN IS MADE IN THE TIP AND IS NEEDED TO STIMULATE CELL ELONGATION AND THEREFORE GROWTH. WITHOUT THE TIP, AND THEREFORE WITHOUT AUXIN, THERE IS NO FURTHER GROWTH IN	THIS PROVES THAT IT IS THE EFFECT OF LIGHT ON AUXIN THAT CAUSES PHOTOTROPISM. THERE IS NO UNEVEN DISTRIBUTION OF AUXIN THROUGHOUT THE SHOOT TIP, SO GROWTH IS EVEN	THIS PROVES THAT SHOOTS GROW TOWARDS THE LIGHT BECAUSE THERE IS AN UNEVEN DISTRIBUTION OF AUXIN IN THE SHOOT TIP, CAUSED BY AUXIN'S RESPONSE TO LIGHT